



University Of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination – Semester I – UCSC AY22 [held in September 2025]

SCS1305 – Computer Systems

(Two (2) Hours)

103

Answer ALL questions

Number of Pages = 12

Number of Questions = 4

To be completed by the candidate

Index No:

Important Instructions to candidates:

1. The medium of instructions and questions is **English**.
2. Note that questions appear on both sides of the paper.
3. If a page or part of this paper is not printed, please inform the supervisor immediately.
4. Write your index number on every page of the Question paper.
5. Answer **ALL** questions.
6. There are **04** questions in **12** pages.
7. All **04** questions carry **equal marks**.
8. Write your answers and index number **clearly and readable** manner.
9. In questions where justifications, reasons, or steps of calculations are specifically asked, the student must provide the same in the answer. **Failure to provide** the above as required in the question will result in that question or part being awarded **zero marks**.
10. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
11. Calculators are **NOT allowed**.

For Examiner's use only

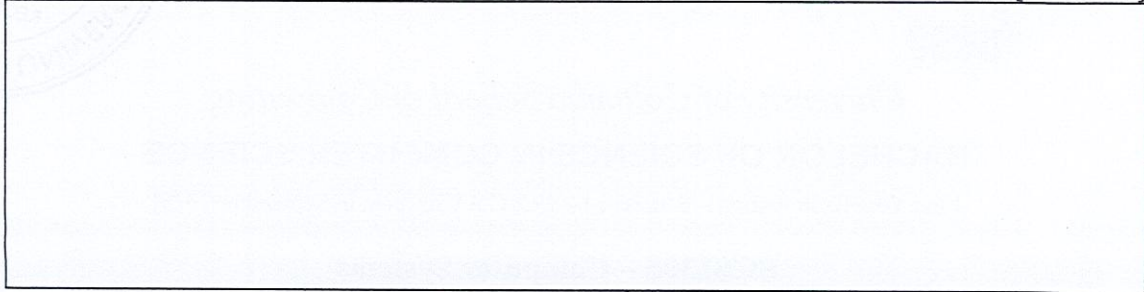
Question No	Marks
1	
2	
3	
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Total	

Index No:

Question 1

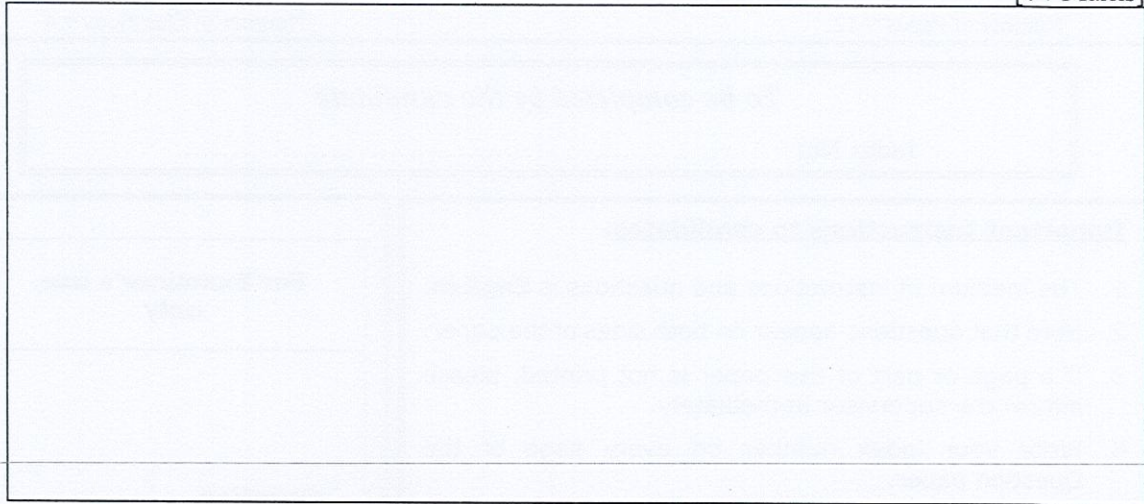
- A. State **two (02)** decimal numbers whose representations are **identical** in *binary*, *octal*, and *hexadecimal* number systems.

[03 Marks]



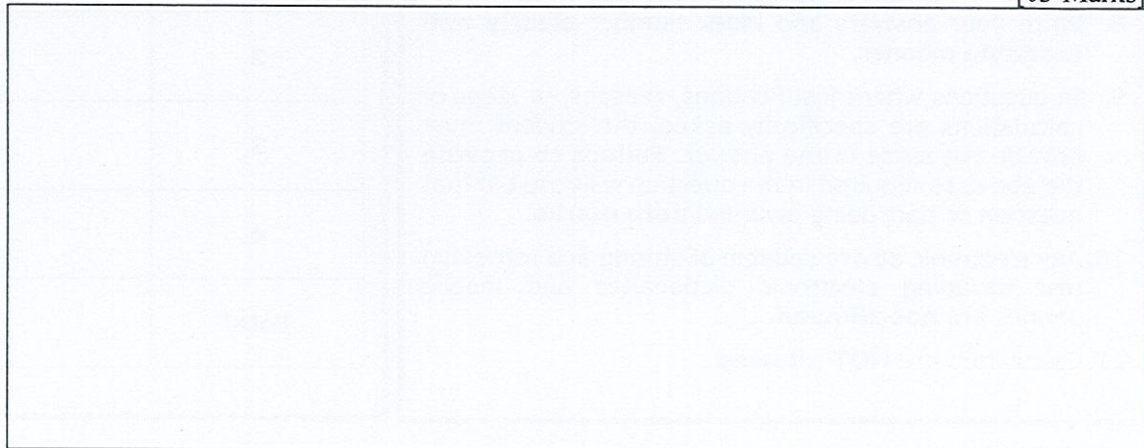
- B. Determine the **minimum** number of bits required to represent the integer range from **-32,767** to **+32,767** in *signed magnitude* representation. Clearly justify your answer.

[04 Marks]



- C. Convert the *hexadecimal* number **80E3** into its equivalent *binary* representation. Show the steps clearly.

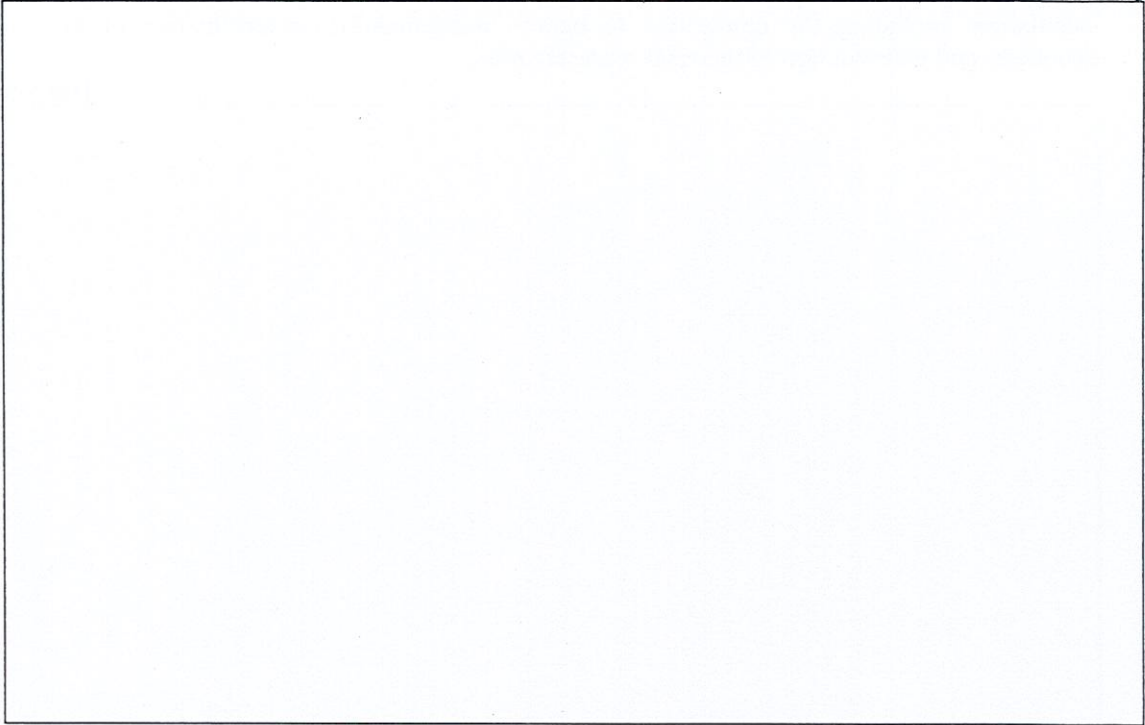
[03 Marks]



Index No:

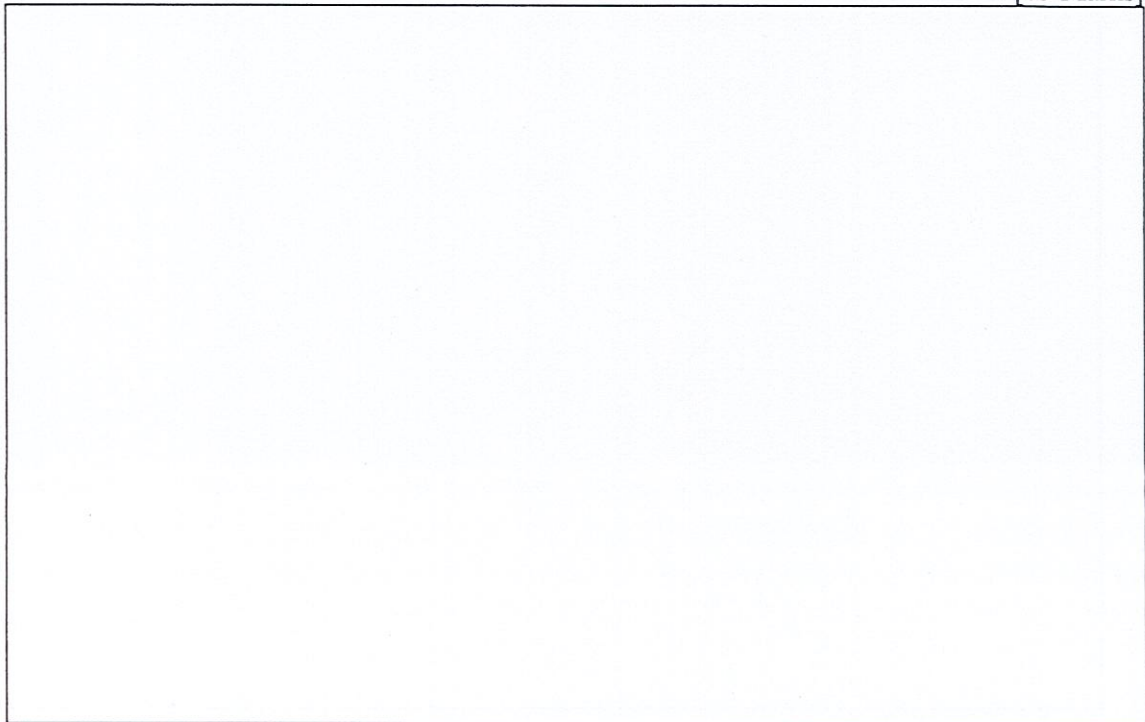
- D. Determine the **maximum** and **minimum decimal values** that can be represented in 16-bit **two's complement** representation. Show your steps and reasoning.

[04 Marks]



- E. Convert the *binary* number 1101.011 into its equivalent *decimal* representation. Show all necessary steps in your calculation.

[05 Marks]



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- F. In the *IEEE 754 single-precision* floating-point format, 1 bit is used for the sign, 8 bits for the exponent (with an *excess-127* bias), and 23 bits for the significand (mantissa). Determine the *IEEE 754 single-precision* representation of the *decimal* number **13.375**. Show all steps in your calculation, including the conversion to binary, normalization, determination of the biased exponent, and construction of the final representation.

[06 Marks]

Question 2

- A. Simplify the following Boolean function F to its simplest form using Boolean algebraic laws, rules, and theorems. Show intermediate steps clearly.

$$F(A, B, C) = (\bar{A} + B)(A + \bar{B}) + \bar{A}.B.\bar{C} + A.\bar{B}.C$$

[05 Marks]

- B. Consider the Boolean function $F(a, b, c, d) = \sum m(5, 6, 7, 13, 14)$, given in *sum-of-minterms* form. The relevant *don't-care conditions* for the function are specified as $\sum m(0, 4, 11, 15)$.

[06 Marks]

- i. Fill in the provided Karnaugh Map (K-Map) with **1s** for the *minterms* of the function $F(a, b, c, d)$, **Xs** for the *don't-care* terms, and **0s** otherwise. Pay careful attention to the column and row labels before filling the K-Map. [03 Marks]
- ii. Using the K-Map, derive the simplified Boolean expression for $F(a, b, c, d)$, by clearly showing the formation of the fewest possible groups. [03 Marks]

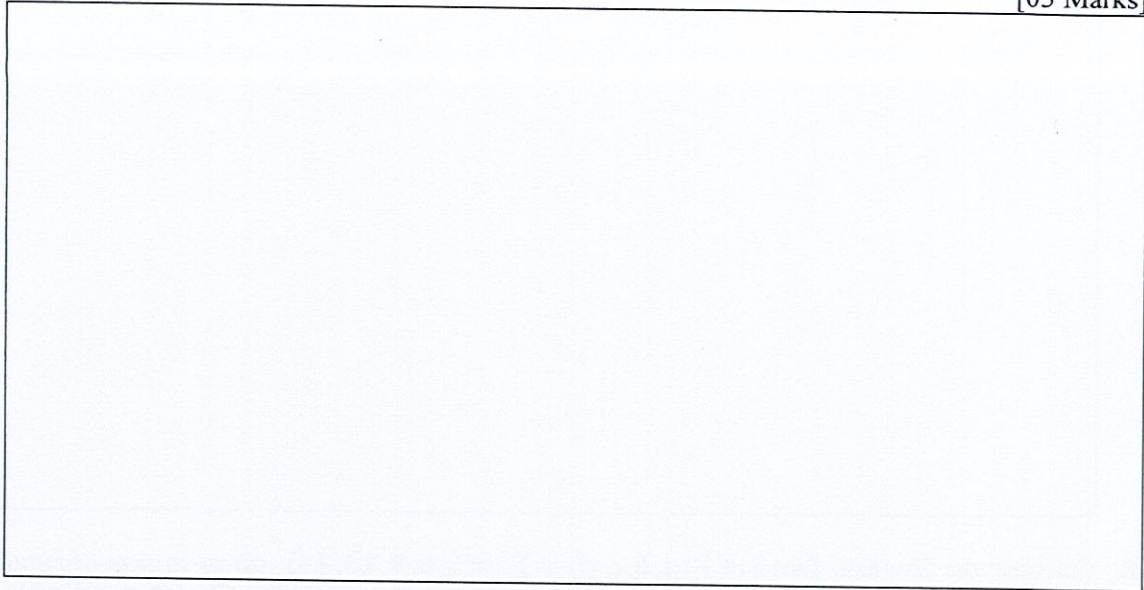
	$a.b$	$a.b'$	$a'.b'$	$a'.b$	
$c.d$					
$c.d'$					
$c'.d'$					
$c'.d$					

Index No:

- C. Given here is the truth table of the *Half-adder*. Using only NAND gates, draw the logic circuit implementation of the *Half-Adder* in the below given space. Clearly label all inputs and outputs.

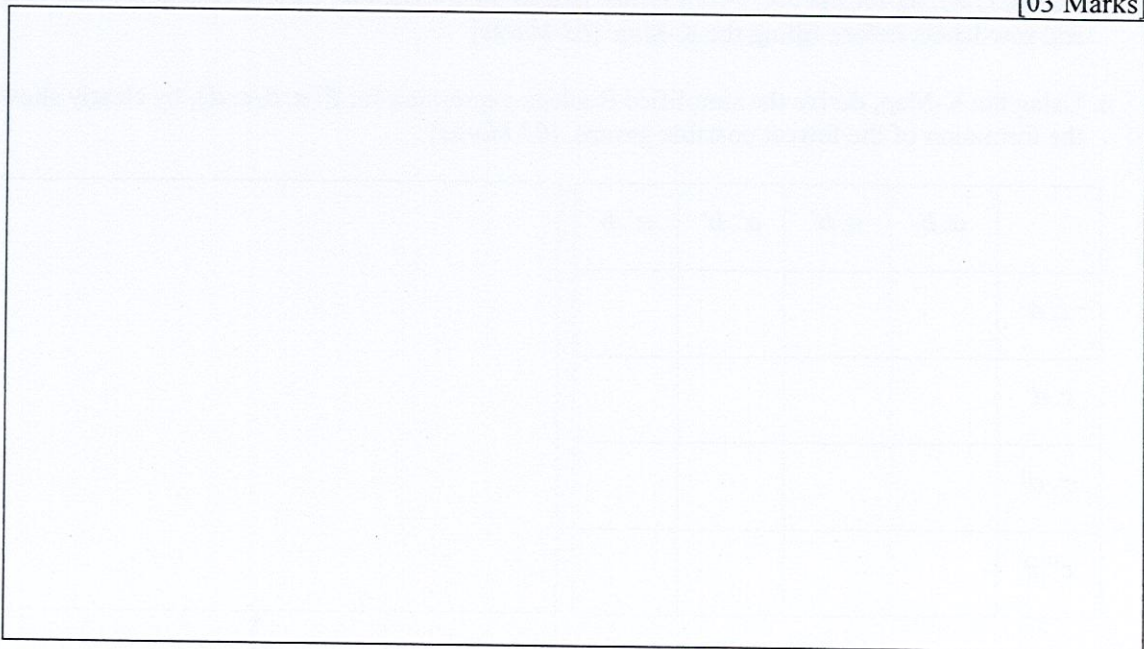
Inputs		Outputs	
A	B	Sum	Carry Out
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

[05 Marks]



- D. State **three (03)** reasons why it is important to simplify Boolean expressions in digital logic circuit design. Briefly justify each reason.

[03 Marks]



- E. Consider the sequential logic circuit given in Figure 2.1, which consist of **positive edge triggered D flip flops**. The external clock input is given as ' Clk_0 ' in the diagram.

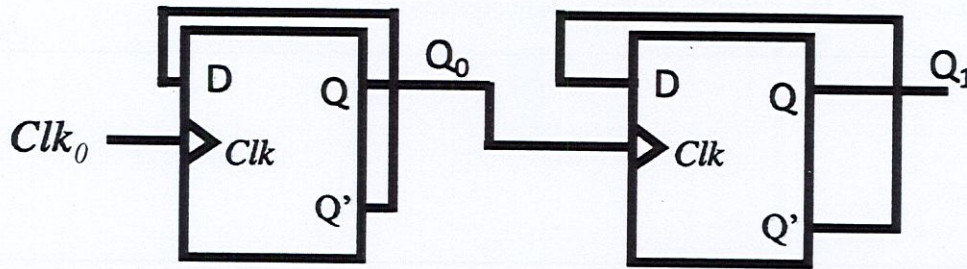
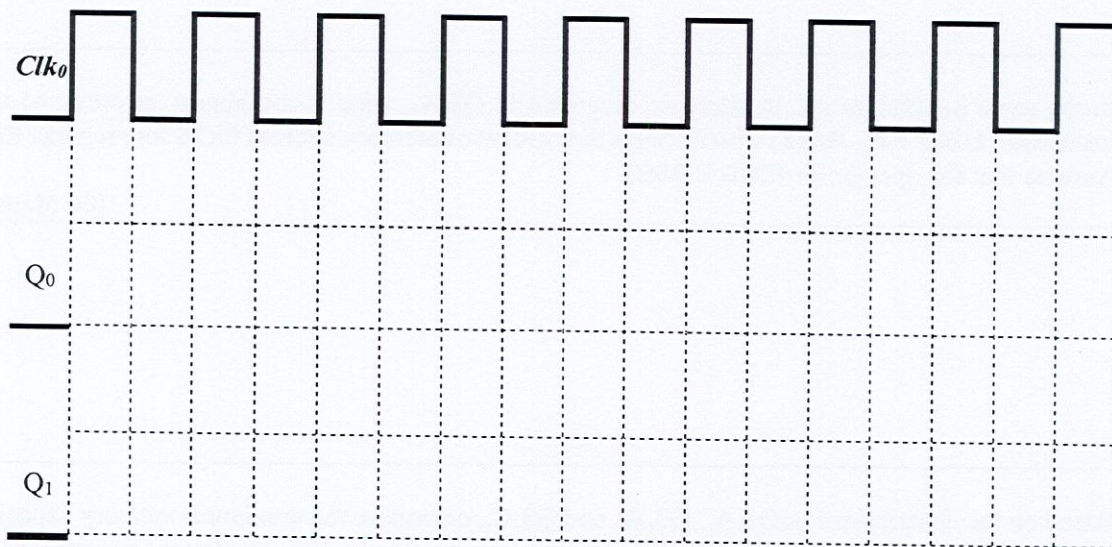


Figure 2.1 – The circuit diagram of the sequential logic circuit.

Complete the timing diagram given below for the above sequential logic circuit. Assume both Q_0 and Q_1 are in logic value zero (0) at the beginning as given in the timing diagram.

[06 Marks]



Question 3

- A. Consider a CPU architecture that uses 16 bits fixed-length instructions and addresses its registers using 4 bits. Registers are directly addressed by using respective register identifiers. Encode the instruction **ADD R0, R1, R2** in binary, assuming the opcode length is fixed at 4 bits and Opcode for **ADD** is **0100**. The memory system is *byte addressable*.

[04 Marks]

- B. What is the maximum number of general-purpose registers that the architecture described in **Q3.A.** can support? Show your steps and calculations to justify your answer.

[03 Marks]

- C. In the same instruction set architecture described in **Q3.A.**, what is the binary encoding of the instruction **LOAD R1, 0x16**, which copies the content of memory address **0x16** into register **R1**? Assume that the opcode for **LOAD** is **0001**.

[04 Marks]

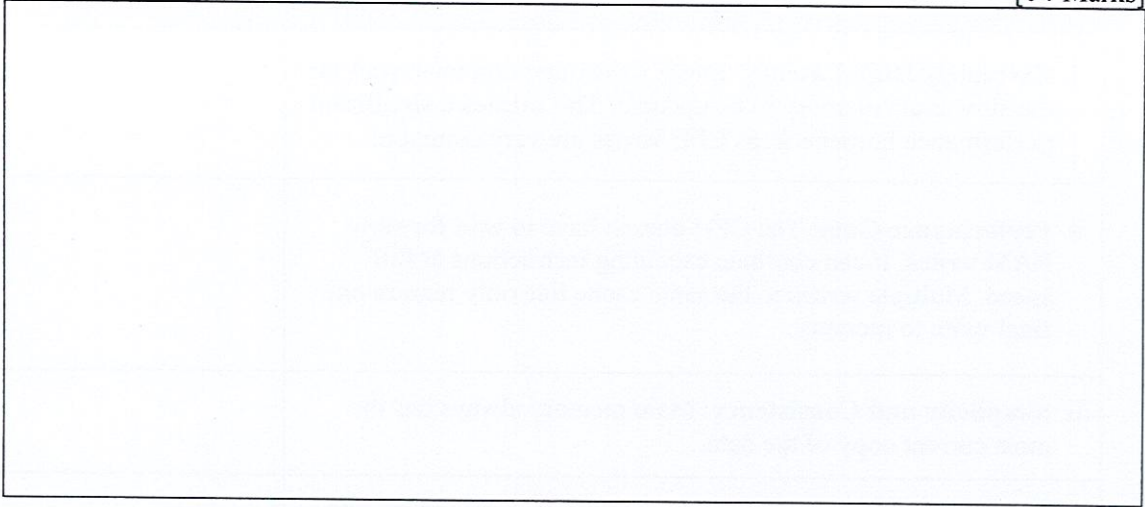
- D. Based on the descriptions in **Q3.A.**, **Q3.B.**, and **Q3.C.**, determine the maximum memory capacity this system supports (Assume a byte-addressable memory). Show your calculations to justify your answer.

[04 Marks]

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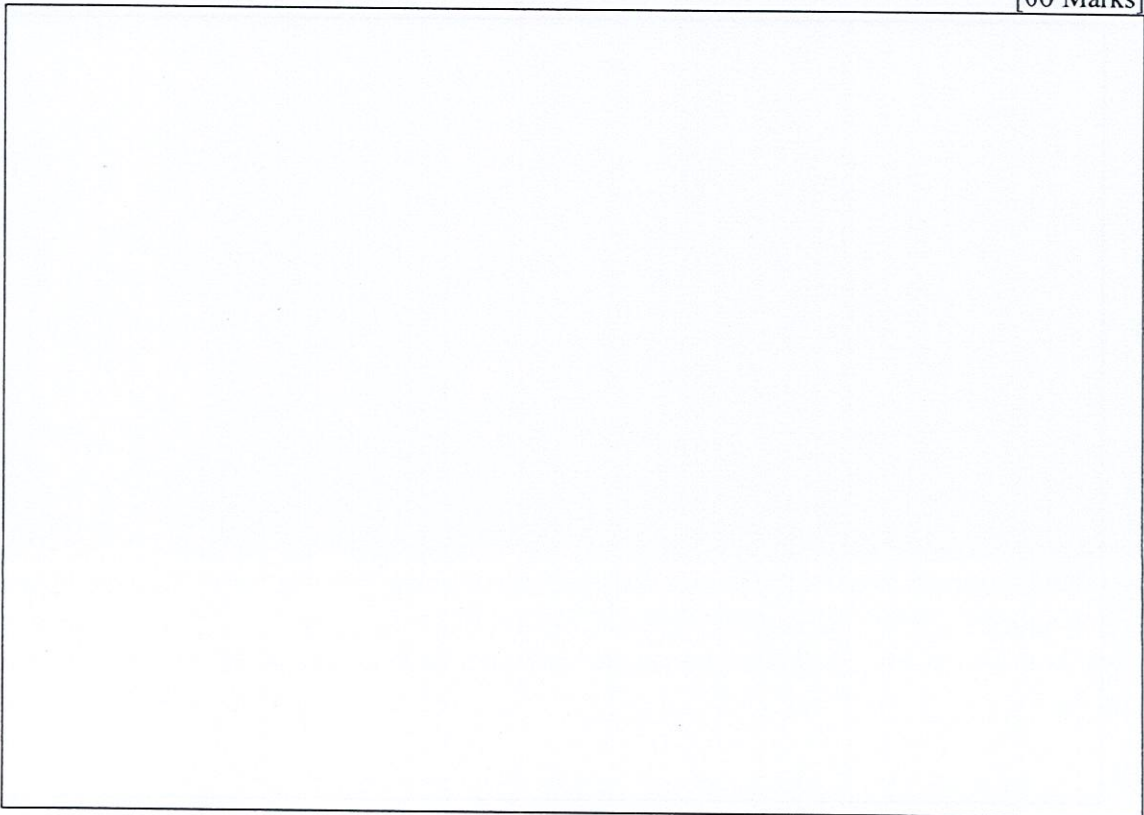
- E. A CPU operates at a clock frequency of 400 MHz. Calculate the corresponding clock cycle time of the CPU in seconds. Show all steps in your calculation and clearly explain the relationship between clock frequency and clock cycle time.

[04 Marks]



- F. Assume two computers, *Machine A* and *Machine B*, implement the same *Instruction Set Architecture* (ISA) but differ in their hardware implementations. *Machine A* has a *clock cycle time* of 12 nanoseconds and a *CPI* (Cycles Per Instruction) of 1.4 for program *P*. *Machine B* has a *clock cycle time* of 14 nanoseconds and a *CPI* of 1.2 for the same program. Which machine executes program *P* faster, and why?

[06 Marks]



Question 4

A. State which cache *write policy* matches with the each of the following characteristics.

[04 Marks]

Characteristics	Your Answer
i. Extremely High Latency: Every write operation must wait for the slow main memory to be updated. This creates a significant performance bottleneck, as CPU writes are very common.	
ii. Performance Gain: The CPU doesn't have to wait for slow RAM writes. It can continue executing instructions at full speed. Multiple writes to the same cache line only require one final write to memory.	
iii. Simplicity and Consistency: Main memory always has the most current copy of the data.	
iv. Complexity: The cache controller must maintain a "dirty bit" for each cache line to track what needs to be written back.	

B. Consider a *byte-addressable* computer system with a 32 KB *direct-mapped* cache that uses 64-byte *cache blocks*. The memory address length is 32 bits. Calculate the number of *offset* bits, *index* bits, and *tag* bits required for the cache system. Show your calculations and steps clearly.

[05 Marks]

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- C. Consider an 8-way set-associative cache in a word-addressable memory system with the word size of 32 bits. The cache has a total size of 4 MB and uses 16 KB of cache blocks. The memory address length is 32 bits. Calculate the number of *offset* bits, *index* bits, and *tag* bits of the cache system. Show your calculations and steps clearly.

[05 Marks]

- D. Describe how the CPU locates a block in the cache described in Q4.C., using the address 0x84201235. Clearly show the *tag*, *index*, and *offset* bits derived from this address.

[05 Marks]

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E. Assume that six different cache blocks, A, B, C, D, E, and F, are mapped to the same cache set in a 4-way set-associative cache. The CPU accesses these blocks in the following order:

- A, B, C, D, A, E, C, A, F

Assuming the cache is initially empty, show the final state of the set separately under both *LRU* (Least Recently Used) and *FIFO* (First-In First-Out) replacement policies by clearly showing each step of the process taken in both scenarios. Write your answers in the relevant cage given below.

[06 Marks]

Least Recently Used

First-In First-Out
